

Polish Spaces and the Borel Hierarchy

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1 Introduction

In this essay, we aim to give a gentle introduction to Descriptive Set theory, primarily by examining the fundamental objects that serve as the setting for all that follows. Descriptive Set theory can be described as the study of “well-behaved” subsets of the real line and, more generally, of Polish spaces, which will be the focus of the first half of the essay.

The motivation to study Polish spaces comes from the fact that, although the Axiom of Choice lets us construct subsets of the real numbers which are not measurable, in practice in Analysis and Probability theory, the sets that we work with primarily are “well-behaved” and hence we would like to have a good understanding of them specifically.

As the name suggests, the study of Polish spaces emerged and became prominent thanks to Polish mathematicians, during a significant period in Polish mathematics and history during the interwar years (1918-1939). After over a century of partition Poland finally gained independence, and quickly became a prominent centre for mathematical research. It became one of the dimensions that helped form Poland’s national identity and unite people in the time of turmoil[8, 6].

The intellectual foundation for Poland’s new mathematical journey was laid by Zygmunt Janiszewski (1888–1920). Focusing on Set theoretic fields was a strategic choice; one that seems natural to us right now. However, at the time, Set theory had not yet received proper recognition by the mathematical community, which demonstrates Janiszewski’s insight and courage[6].

In this essay, we will examine Polish spaces and their main properties, touching on Borel sets and the Borel hierarchy. We will also look at different examples of Polish spaces being used in other areas of mathematics. For example, Polish spaces are a convenient setting for Measure theory, in particular in Probability theory.

Hopefully, by the end of this essay, the readers will have a foundational grasp of the essence of Descriptive Set theory, or at least the properties and understanding of Polish spaces.

The main resource for this essay is Kechris’s graduate-level textbook[3].

2 Polish Spaces

2.1 Important definitions

We shall begin our exploration by formally defining one of the most fundamental concepts in Descriptive Set theory - what it means for a topological

space to be Polish.

Definition 1 (Polish space). [3, p.13] *We call a topological space Polish if it is a separable completely metrizable topological space.*

As a reminder:

- A metric space M is called *complete* if every Cauchy sequence of points in M converges to a point in M . E.g. the real numbers equipped with the standard metric¹ are complete.
- A topological space is separable if it *contains a countable dense subset*. I.e. if there exists a countably infinite sequence of elements of the space, such that every non-empty open subset of the space contains at least one element of the sequence.

Note the importance of the word “metrizable”. We are specifically looking at those topological spaces whose topology is induced by some metric.

Yet, it is important to choose the metric correctly. For example, $(0, 1)$ with the standard Euclidean metric is not complete, but $(0, 1)$ is still Polish, as we can find a different metric that indeed makes the interval complete. For example, we can take $f(x) = \tan(\pi(x - \frac{1}{2}))$, which is a homeomorphism to $\mathbb{R}$. Pulling back the Euclidean metric on $\mathbb{R}$ will give us the desired metric on $(0, 1)$.

A standard example of a Polish space is the familiar real number line. Let’s look at $\mathbb{R}$ endowed with the typical Euclidean metric, and let’s take $\mathbb{Q}$ as a subset of $\mathbb{R}$. From Analysis 1 we indeed know that any Cauchy sequence of real numbers converges, so $\mathbb{R}$ is complete. Moreover, $\mathbb{Q}$ is a countable dense subset of $\mathbb{R}$ as any open interval (a, b) contains a rational number.

It may be intuitive to rephrase the concept of denseness here as “it’s possible to get arbitrarily close to any real number using only rational numbers”.

It might also be helpful to look at a non-example: the rationals themselves endowed with the Euclidean metric do not form a Polish Space, as a Cauchy sequence of rationals may have an irrational limit point, so $(\mathbb{Q}, \|x - y\|_2)$ is not complete². In fact, it turns out *there does not exist a complete metric*

¹As we will see further on, specifying the metric used is quite important. It also makes sense – we are talking about “closeness” of points, so it is important how we define the distance between them.

²Note, that $\mathbb{Q}$ is an example of a metrizable non-complete space. One might wonder whether there exists a complete space which is not metrizable – and the common-sense answer will be that the question doesn’t make sense. Completeness is a property that is downstream of distance – we’re talking about Cauchy sequences, and for that we need a notion of distance! The more peculiar answer is left to the reader to discover.

that induces the standard topology of $\mathbb{Q}$. Let's take some time to understand why this is the case.

We will need to use the result of Corollary 9.25 from the MA260 lecture notes. The Corollary states the following: Let $\{F_j\}_{j=1}^\infty$ be a countable collection of nowhere dense subsets of a non-empty complete metric space (X, d) . Then $\bigcup_{j=1}^\infty F_j \neq X$.

Lemma 1. $\mathbb{Q}$ is not completely metrizable.

Proof. Assume $\mathbb{Q}$ is actually completely metrizable. Then there exists a metric d such that $(\mathbb{Q}, d)$ is a non-empty complete metric space. We know $\mathbb{Q}$ is countable, so we can enumerate all the rationals to get $\{\{q_1\}, \{q_2\}, \{q_3\}, \dots\}$, a countable collection of singleton subsets of $(\mathbb{Q}, d)$. Moreover, as each $\{q_i\}$ is a singleton set, it is nowhere dense under the standard topology of $\mathbb{Q}$. But then $\bigcup_{i=1}^\infty \{q_i\} = \mathbb{Q}$ holds and is a contradiction to Corollary 9.25. $\square$

From the definition of a Polish Space it naturally follows that all Polish spaces are second-countable.

Definition 2 (Second countable). [7, p.190] *A topological space X is called second-countable if it has a countable basis for its topology.*

In practice, this means that there exists some countable collection U of open subsets of X such that any open subset of X can be written as a union of members of U .

So even if the space isn't conventionally countable, having a countable base is the next best thing we can get in regards to countability. Having the property of second-countability is important for various lemmas and objects later on, and deserves to be examined in some detail.

Lemma 2. [2, p.148] *Every Polish space is second-countable.*

Proof. Let X be our Polish space, and let $D = d_1, d_2, \dots$ be a countable dense subset of X . Let's show that $B = B(d_i, 1/n) : d_i \in D, n \in \mathbb{N}$ is a countable base of open balls. It's quite easy to see that this set is countable, as it is indexed by two countable indices, and the product of a finite number of countable sets is countable. Now, to see that this is a basis, take any open set $U \subset X$ and any $x \in U$. As U is open $\exists B(x, \varepsilon) \subset U$. Take n large enough that $\frac{1}{n} < \frac{\varepsilon}{2}$. Now, as D is dense, $\exists d_i : d(x, d_i) < \frac{1}{n}$, i.e. $x \in B(d_i, 1/n)$. Finally, to show that $B(d_i, 1/n)$ is a subset³ of U , take any $y \in B(d_i, \frac{1}{n})$. By the triangle inequality $d(x, y) \leq d(x, d_i) + d(y, d_i) < \frac{1}{n} + \frac{1}{n} < \varepsilon$, i.e. $y \in B(x, \varepsilon) \subset U$. Hence, every element of U is in some basis element, and the basis elements are all inside U , so U is indeed a union of basis elements. $\square$

³We need to do this, otherwise it's unclear why the basis elements don't "overspill" and their union isn't greater than U .

Polish spaces being second-countable means that for example Lindelof's Lemma [4, p. 49] can be applied.

Lemma 3 (Lindelof's Lemma). *Every open cover of a second-countable space has a countable subcover.*

Basically, this lemma lets us go from an arbitrary open cover that is potentially hard to work with, to a "tame" one, which is much easier to comprehend. We understand how to work with sequences, and hence the complexity of the setup reduces greatly compared to the counterfactual. We're almost getting the same perks as working in a compact space, except we don't have finiteness.

Some well-known spaces, which are Polish, include [3, pp.13-14]:

- Familiar intervals on $\mathbb{R}$: the open and closed intervals $(0, 1)$ (as we already saw, depending on the metric) and $[0, 1]$, $\mathbb{R}$. Other familiar objects like the unit circle, $\mathbb{C}$ and $\mathbb{C}^n$ are all Polish. If the metric isn't explicitly mentioned, we shall assume that the standard one is used.

Some slightly more peculiar spaces which are Polish, are for example:

- The Cantor space, $C := 2^{\mathbb{N}}$ is a Polish space, as the product of infinitely many copies of $\{0, 1\}$ with the discrete topology.
- Similarly, the Baire space, $N := \mathbb{N}^{\mathbb{N}}$ is a Polish space.

We will come across the Cantor space later on in this essay, so let's briefly make sure it is indeed Polish. Let's define our metric: $\forall x, y \in 2^{\mathbb{N}} d(x, y) := 2^{-m(x, y)}$, where $m(x, y)$ denotes the first position at which x and y disagree⁴. If $x = y$, we define $m(x, x) := 0$. The space is complete, as elements of a Cauchy sequence at some point start agreeing on longer and longer segments, so every position at some point will be determined. And finally, it's separable, as the sequences with finitely many 1s (i.e. only have 0 after some position) form a dense subset.

Looking at the two spaces above, one might wonder if we can take an infinite product of any other set. And indeed, any space $A^{\mathbb{N}}$ where A is countable with discrete topology is Polish [3, p. 13].

Now that we know what we're working with, let's define another crucial notion and start looking at more interesting properties that add color to everything we have seen so far.

⁴We will leave it up to the reader to check that this is indeed a metric.

2.2 G_δ sets and properties of Polish spaces

One question that might interest us in the context of Polish spaces is: when is a subset of a Polish space also Polish? And the answer to this question is pretty elegant, but we will need to define what it means to be a G_δ set before we proceed⁵.

Definition 3 (G_δ set). [7, p.249] *A subset of a topological space is said to be a G_δ set if it is a countable intersection of open sets*⁶.

G_δ sets are pretty exciting - it's nice to work with countable⁷ sets of objects, as there is a bijection with the naturals and so we can easily index the elements of our set. And it's also nice to work with intersections - we end up only with those elements, which are definitely in *all* the objects we're intersecting. And for G_δ sets we have both of these!

You may think G_δ sets are a completely new concept we have not dealt with at all before - this is actually not the case, they have secretly been lurking in most of the spaces we've seen in lectures!

Lemma 4 (Closed sets are G_δ). [3, p.68] *If X is metrizable, then every closed set is a G_δ set.*

This is pretty easy to prove by taking $\frac{1}{n}, n \in \mathbb{N}$ neighbourhoods of the fixed closed set.

Moreover, any open set is trivially G_δ by taking the open set countably many times.

Equipped with this definition we can answer some interesting questions, for example...

2.2.1 When is a subspace Polish?

To answer this question we will need the following theorem:

Theorem 1 (Kuratowski's Theorem). [3, p.16] *Let X be metrizable, Y be completely metrizable, $A \subseteq X$, and $f : A \rightarrow Y$ be continuous. Then there is a G_δ set G with $A \subseteq G \subseteq \overline{A}$ and a continuous extension $g : G \rightarrow Y$ of f .*

⁵If the reader ever wondered in MA260 what happens if we take countable intersections of open sets instead of only finite ones - here's your moment.

⁶As in the case of open sets, the complements of G_δ sets have a name of their own, although lesser known - F_σ sets. As some might have guessed, F_σ sets are countable unions of closed sets.

⁷Countability is nice for many reasons. It's a generalisation of finiteness that still behaves nicely in some cases. E.g. countable unions of countable sets are countable, the sum of a countable number of positive numbers can be finite (while this is not the case for uncountable sets), etc.

The proof is a little bit out of the scope of this essay, however, it is worth mentioning that an idea we see in the proof [3, p.16], is actually of some familiarity to us and is a generalisation of continuity from first-year analysis: we look at smaller and smaller neighbourhoods of a point, so that the values of the function on those neighbourhoods fluctuate less and less - exactly what we saw in the case of continuity! Moreover, points with “fluctuation” equal to 0 are, unsurprisingly, exactly the points of continuity of the given function.

Now, equipped with Kuratowski’s theorem we are ready to prove the following fact:

Theorem 2. [3, p.17] *A subspace of a Polish space is Polish iff it is G_δ .*

Proof. [3, p.17] Let X be the given Polish Space, and $Y \subset X$. For the first assertion, suppose Y is Polish and consider the identity $id_Y : Y \rightarrow Y$. Trivially, the identity is continuous, so by Kuratowski’s Theorem, there is a G_δ set G with $Y \subseteq G \subseteq \bar{Y}$, where $\bar{Y}$ is the closure of Y in X , and a continuous extension $g : G \rightarrow Y$ of id_Y . As G is a subset of $\bar{Y}$, it means that Y is dense in G . Let us show that $g = id_G$. Indeed, density gives us that $\forall x \in G, \exists (y_n) \in Y$ such that $(y_n) \rightarrow x$. By continuity of g we have $g(x) = g(\lim_{n \rightarrow \infty} y_n) = \lim_{n \rightarrow \infty} g(y_n) = \lim_{n \rightarrow \infty} id_Y(y_n) = \lim_{n \rightarrow \infty} (y_n) = x$. Hence $Y = G$, and so Y is indeed G_δ .

For the opposite direction, write $Y = \bigcap_n U_n$, where U_n is open in X . Define $F_n = X \setminus U_n$.⁸ Let d be a complete metric which induces the topology on X . Define a metric d' for Y as

$$d'(x, y) = d(x, y) + \sum_{n=0}^{\infty} \min \left\{ \frac{1}{2^{n+1}}, \left| \frac{1}{d(x, F_n)} - \frac{1}{d(y, F_n)} \right| \right\}.$$

Note, that this metric is well-defined as⁹ $\frac{1}{d(x, F_n)} > 0$ and the second summand is bounded above by a converging geometric series. Similarly, the two metrics have the same convergent sequences and hence the same topology. We are left to show that (Y, d') is complete¹⁰. Let (y_i) be a Cauchy sequence in (Y, d') . Then it is Cauchy in (X, d) because $d'(x, y) \geq d(x, y)$. So $\exists y \in X$ such that $y_i \rightarrow y$. But note also, as the sequence is Cauchy, for each n , $\lim_{i, j \rightarrow \infty} \left| \frac{1}{d(y_i, F_n)} - \frac{1}{d(y_j, F_n)} \right| = 0$, so for each n , $\frac{1}{d(y_i, F_n)}$ is also Cauchy and converges. It then follows that $d(y_i, F_n)$ is bounded away from 0. As we have $d(y_i, F_n) \rightarrow d(y, F_n)$, we get $d(y, F_n) \neq 0$ for all n , so $y \notin F_n$ for all n , i.e. $y \in Y$. Hence, we get that $y_i \rightarrow y$, and $y \in Y$, so (Y, d') is indeed complete,

⁸Clearly, F_n is closed.

⁹The inequality holds because F_n is closed and x is in Y , so it’s not in F_n !

¹⁰Remember that separability is inherited from X .

and hence Y is completely metrizable. Separability of Y easily follows from the fact that subspaces of second-countable spaces¹¹ are second-countable [7, p.191] and for metrizable spaces being second-countable and separable are equivalent [7, p.192]. $\square$

Immediately from the proof and from Lemma 4, we get that all closed subsets of a Polish space are Polish - we now know a lot about a family of the subspaces of our Polish space basically for free. Moreover, any other results we prove about Polish spaces will also hold for all G_δ subsets, including all the closed subsets.

Now let's look at some more properties of Polish spaces and the relationships between them.

2.3 Polish spaces and well-known spaces

What is the relationship between a general Polish space and two specific and well-studied Polish spaces - the Hilbert cube and the Cantor space? In this section we will have a look at two interesting theorems. These theorems work for more general spaces (one theorem requires a metric space to be separable, and the other - compact), but they are commonly mentioned specifically in the context of Polish spaces as they tie in nicely into the theory of these spaces and how all the members of the family of Polish spaces share “nice” properties among each other.

Let's actually first define the Hilbert cube. There are several specific objects one may mean when talking about the Hilbert cube:

Definition 4 (Hilbert cube). [11, p.65] *The Hilbert cube is the topological product of the intervals $[0, 1/n]$ for $n = 1, 2, 3, \dots$*

We will primarily be interested in a version of it denoted by $\mathbb{I}^\mathbb{N}$, which is the topological product $\prod_{n=1}^\infty [0, 1]$.

It's quite easy to see that the two versions are homeomorphic to each other, so if a space is homeomorphic to one of these, it automatically implies that there exists homeomorphism to the other.

One might wonder what's the point of having the $[0, 1/n]$ version of the Hilbert cube - it's useful when embedding the Hilbert cube into l^2 , because then every point satisfying these bounds is in l^2 .

Now that we have our object of interest defined, we can proceed.

¹¹ X is second-countable by Lemma 2.

2.3.1 Homeomorphism

Theorem 3 (Homeomorphism property). [3, p. 22] *Every separable metric space is homeomorphic to a subspace of the Hilbert cube $\mathbb{I}^{\mathbb{N}}$.*

Proof. [3, p. 22] Let (X, d) be a separable metric space, with¹² $d \leq 1$. As it is separable, we can take (x_n) which is dense in X . Define $f(x) = (d(x, x_n))_{n=1,2,\dots}$. (This is exactly why we need $d \leq 1$, so that the image lands inside the cube). As d is a metric and (x_n) is dense, f is clearly continuous and injective. We are left to show that $f^{-1} : f(X) \rightarrow X$ is continuous. Let $f(a_m) \rightarrow f(a)$, i.e. $d(a_m, x_n) \rightarrow d(a, x_n) \forall n$. Fix $\varepsilon > 0$ and let n be such that¹³ $d(a, x_n) < \varepsilon$. Now, since $d(a_m, x_n) \rightarrow d(a, x_n)$, $\exists M$ such that: $m \geq M \Rightarrow d(a_m, x_n) < \varepsilon$. Then by the triangle inequality $d(a_m, a) < 2\varepsilon$, i.e. $a_m \rightarrow a$, so f^{-1} is indeed continuous, and so f is a homeomorphism. $\square$

If one observes closely, it follows that “Polish spaces are, up to a homeomorphism, exactly the G_δ subspaces of the Hilbert cube” [3, p. 22]. Indeed, Theorem 2 told us that G_δ subspaces of Polish spaces – the Hilbert cube is, of course, Polish – are Polish. And if a Polish space isn’t a subspace of the Hilbert cube already, by Theorem 3 a homeomorphism embeds it into some subspace of the Hilbert cube. As complete metrizability¹⁴ and separability [1, p. 31] are preserved under homeomorphisms, those subspaces are also Polish, and hence G_δ , which proves the statement.

2.3.2 Continuous image of Cantor space

Now that we understood the relationship between a general Polish space and the Hilbert cube, let us move on to see how the Cantor space ties into this. Note that we are introducing another requirement for our space - compactness.

Theorem 4 (Continuous image). [3, p. 23] *Every non-empty compact metrizable space is a continuous image of the Cantor space C .*

¹²If our initial space has d not bounded by 1, then we can replace it with an equivalent metric without changing the topology, e.g. $d' = \min(d(x, y), 1)$.

¹³This is possible as (x_n) is dense.

¹⁴We have seen without proof that being metrizable is a topological property in MA260. A nice proof of this is given in Ivan Khatchaturian’s lecture notes [5, pp. 5]. Let’s quickly show that if (X, d) is a completely metrizable metric space, and $f : X \rightarrow Y$ is a homeomorphism, then Y is also completely metrizable. We will use the metric $\rho(y_1, y_2) := d(f^{-1}(y_1), f^{-1}(y_2))$ on Y . The lecture notes show that ρ already generates the correct topology on Y . If (y_n) is Cauchy, then $(f^{-1}(y_n))$ is Cauchy. Then $f^{-1}(y_n) \rightarrow x, x \in X$. But then, as f is continuous, $y_n \rightarrow f(x)$. $f(x) \in Y$, which proves the claim.

Proof. [3, p. 23] First, let us show that $\mathbb{I}^{\mathbb{N}}$ is a continuous image of C . Define $f(x) : C \rightarrow \mathbb{I}$ to be $f(x) = \sum_{n=0}^{\infty} x(n)2^{-n-1}$, where $x(n)$ denotes the n -th coordinate of x . Clearly each partial sum $f_N(x)$ of f is continuous. These partial sums uniformly converge to f , and so f is also continuous. Applying f coordinate-wise, we get a continuous mapping of $C^{\mathbb{N}}$ onto $\mathbb{I}^{\mathbb{N}}$. The desired result is obtained by noting that $C^{\mathbb{N}}$ is homeomorphic to C .

Now, because compact metrizable spaces are separable [1, p. 256], we can use Theorem 3. Hence, every compact metrizable space X will embed into a compact subset of $\mathbb{I}^{\mathbb{N}}$. Compact subsets of Hausdorff spaces are closed, and hence the preimage, call it F , of this subset is also closed in C . Hence, we have a continuous surjection $g : F \rightarrow X$.

The final step, used by both Kechris and Engelking [1, p. 71], is to say that there exists a continuous surjection $h : C \rightarrow F$, such that $\forall x \in F : h(x) = x$, [3, p. 8]. Applying $g \circ h$ to C will give us X as the continuous image. $\square$

Let's take a moment to understand why it was important to introduce the requirement of compactness for X . The final step with the desired surjection from C to F works only if F is closed. We obtained F as a preimage of a subset of $\mathbb{I}^{\mathbb{N}}$, and an easy way to guarantee it is closed was to have it be the preimage of a closed set. If X were not compact, its embedding in $\mathbb{I}^{\mathbb{N}}$ need not have been closed.

Combining Theorem 3 and Theorem 4, we obtain that compact Polish spaces have the unique property that both universality properties hold for them. This gives a very interesting duality of perspectives - Theorem 3 gives us the direction from a compact Polish space into a “well-understood explicit object”, and Theorem 4 gives us the direction from a “well-understood explicit object” into a compact Polish space.

So far, to prove all the properties, we have only used, quite expectedly, well-known techniques from Analysis, Topology, etc. However, as we are about to see, Game theory also has a role to play in the context of Descriptive Set theory.

2.4 Choquet's game

Before we begin - let's draw our attention to a little historical note, which once again makes me, and I hope the readers as well, appreciate how the results we take for granted nowadays took years and multiple contributors to develop, generalize and refine.

The story of Choquet's game actually started with the Polish mathematician S. Mazur, who proposed a 2-person game on the real line. As folklore goes, the problem was solved by none other than St. Banach in 1935, but

the proof was never published. More than 20 years later, in 1957, J. Oxtoby published a proof for general topological spaces. Finally, the game reached Choquet, who turned the game's attention to the topological space itself, while the previous iterations focused¹⁵ on characterising a given subset [9, pp. 3-4].

Definition 5 (Strong Choquet Game). [3, pp. 44-45] *Given a non-empty topological space X , the strong Choquet game G_X^s is defined as follows: Players I and II take turns in playing non-empty open subsets of X (U_i for Player I and V_i for Player II), and also I is required to play a point $x_i \in U_i$ so that:*

- $\forall i$ we have $U_i \supset V_i$, and $V_i \supset U_{i+1}$
- $x_i \in V_i$ as well, i.e. x_i is an additional constraint on Player II.

Player II wins this game if $\bigcap_n V_n (= \bigcap_n U_n) \neq \emptyset$. (Hence, Player I wins if $\bigcap_n V_n (= \bigcap_n U_n) = \emptyset$).

Definition 6 (Strong Choquet space). [3, p. 45] *A nonempty space X is said to be strong Choquet if Player II has a winning strategy in G_X^s .*

The characterisation which follows from this game creates an elegant connection between Game Theory and Descriptive Set Theory.

Theorem 5 (Choquet's Characterization). [3, p. 45] *Let X be a nonempty separable metrizable space dense in a Polish space P .*

Then X is strong Choquet $\leftrightarrow X$ is Polish.

The proof of Choquet's Characterization is out of the scope of this essay, but involves some very interesting strict formalisation of the notions of "strategy" and relies on the tree of possible and allowed positions. The reader is highly encouraged to have an independent read in Kechris's book [3, pp. 43-45].

Intuitively, we can see that the characterization works, as the optimal strategy for the second player is to construct a convergent Cauchy sequence. Hence, if the limit point is in the space, indeed we will have the intersection of all V_n be non-empty and containing this point.

¹⁵The Banach-Mazur game and Oxtoby's generalization focused on whether a given subset E was meagre (i.e. was a countable union of subsets which are not dense in any non-empty open set.)

2.5 Applications of Polish Spaces

Polish spaces interested us in the first place because they are a very nice setting for functions, and much of Analysis is naturally done on Polish spaces.

It comes as no surprise that Polish spaces are of a lot of interest in Measure theory, and especially in Probability theory, as they guarantee nice and expected behaviour of most of the machinery used.

A somewhat entertaining example is known as Nedoma's pathology [10, p. 288]. It turns out that if you take a measurable space $(X, \mathcal{A})$ with the cardinality of X greater than the continuum, then the diagonal of $X \times X$ can never be in the product σ -algebra, i.e., is not a measurable set. This is a problem, as we usually want to be able to compare random variables, so if X_1 and X_2 are two random variables taking values in our measure space, then the question "what is the probability of X_1 and X_2 being equal?" is unanswerable. However, uncountable Polish spaces always have the size of the continuum [3, p. 32], so we don't have to worry about this in the usual setting.

3 Borel sets and Borel hierarchy

Having established and explored the properties of these "nice" Polish spaces, a natural question arises: do their subsets form any meaningful structure? As it turns out, for a specific collection of those subsets they do - the pattern is captured by the Borel hierarchy. In this section, we explore its construction and properties.

3.1 Definitions

Definition 7 (σ -algebra). [3, p. 47] *A σ -algebra on a set X is a collection of subsets of X which includes the empty set, is closed under countable unions and is closed under complements.*

Note, that the last two properties combined also give us closure under countable intersections.

For each $U \subseteq \text{Pow}(X)$, there is a smallest σ -algebra containing U . We say that this σ -algebra is "generated by U " and denote it by $\sigma(U)$ [3, p. 65].

Now, we are ready to start putting "Borel" into our definitions.

Definition 8. [3, p. 65] *Given a topological space (X, T) , the class of Borel sets on X is the σ -algebra generated by the open sets of X . We denote it by $B(X)$.*

It follows that the Borel sets of X are the elements of the σ -algebra generated by the open sets of X .

Note, that $B(X)$ can indeed be defined for any space (X, T) , however, if it's Polish we automatically get some nice properties, as we will see a bit later. These definitions are enough for us to proceed to examine the...

3.2 Borel hierarchy!

Suppose we have a metrizable space X , so that every closed set is G_δ ¹⁶.

Recall, at our disposal we have the Borel sets - the elements of the σ -algebra $B(X)$. A natural thing to try and do with them is impose some sort of ordering, for example, by looking at the complexity of their construction. Specifically, we shall look at the number of times we needed to apply countable intersections and unions to obtain our set from just the open sets of X . (Below we will use “set” as a shorthand for “Borel set”.)

We construct the hierarchy inductively[3, p. 68].

- Let $\Sigma_1^0(X)$ be the set of all open sets of X .
- Let Π_ξ^0 be defined as $\sim \Sigma_\xi^0$
- Let Σ_ξ^0 be the collection of those sets, which can be written as a countable union of sets belonging to some Π_α^0 , with $\alpha < \xi$.
- Additionally, we denote by Δ_ξ^0 the intersection $\Pi_\xi^0 \cap \Sigma_\xi^0$.

Note, that ξ here ranges over all countable ordinals, not just the natural numbers. The first uncountable ordinal will be denoted by ω_1 from now on.

Let's try to understand what we constructed. By definition, members of Σ_1^0 are the open sets. Then what are the members of Π_1^0 ? All the sets, which are complements of open sets, i.e. exactly the closed sets. And what can we say about the members of Π_2^0 ? It is easy to see by De Morgan laws that they are exactly the G_δ sets.

Additionally, a little insight into what is to come - notice that¹⁷, as all closed sets of X are G_δ we have $\Pi_1^0 \subseteq \Pi_2^0$. Consider this a little “sanity” check for the assumption we are about to make: in the discussion below, we will take for granted that we have the inclusion $\Sigma_a^0(X) \subset \Sigma_b^0(X)$ for $a < b$, and similarly if we take $\Pi_\xi^0(X)$. We will see this confirmed later on.

¹⁶We will later see why this is an important assumption to have.

¹⁷This is exactly why we wanted X to be metrizable.

Going further into the hierarchy we start losing the ability to clearly define the sets we're looking at. However, there is a way to gain more clarity on their construction.

Denote by $G(X)$ and $F(X)$ the classes of open and closed subsets of X respectively. Additionally, let $U_\sigma = \bigcup_n U_n : U_n \in U$ and $U_\delta = \bigcap_n U_n : U_n \in U$.

Then, $\Sigma_1^0(X) = G(X)$, $\Pi_1^0(X) = F(X)$, $\Sigma_2^0(X) = (F(X))_\sigma = F_\sigma(X)$, by De Morgan laws we have $\Pi_2^0(X) = (G(X))_\delta = G_\delta(X)$, then $\Sigma_3^0(X) = (G_\delta(X))_\sigma = G_{\delta\sigma}(X)$, etc[3, p. 69].

Continuing like this, we have that each level of the hierarchy can be written as either $G_{s_1}(X)$ or $F_{s_2}(X)$, where s_1 is a finite (possibly empty) alteration of σ and δ beginning with δ . And similarly s_2 is a finite (possibly empty) alteration of σ and δ beginning with σ .

Now that we have defined our structure, let's look at some properties which illuminate why it indeed makes sense to call it a "hierarchy"[3, p. 69]. Firstly, $\forall \xi$ Σ_ξ^0 , Π_ξ^0 and Δ_ξ^0 are subsets of each of Σ_β^0 , Π_β^0 and Δ_β^0 , whenever $\beta > \xi$. Next, Σ_ξ^0 and Π_ξ^0 are subsets of $\Delta_{\xi+1}^0$. And finally, $B(X) = \bigcup_{\xi < \omega_1} \Sigma_\xi^0(X) = \bigcup_{\xi < \omega_1} \Pi_\xi^0(X) = \bigcup_{\xi < \omega_1} \Delta_\xi^0(X)$.

We get that $B(X)$ indeed decomposes into layers, moreover, the relationship between the layers indeed forms some sort of one-directional hierarchy.

It's worth noting, that the "contents" of the hierarchy are not universal - it explicitly depends on the underlying topology of X . For example, if we take $\mathbb{R}$ with the Euclidean metric, the only clopen sets are $\mathbb{R}$ and $\emptyset$. However, $\mathbb{Q}$ with the discrete metric has infinitely many clopen sets. Hence, for these two spaces the corresponding Borel hierarchies have radical differences even at the very first level $\Delta_1^0(X)$ of the hierarchy.

3.2.1 But why Polish?

Notice, that when constructing the hierarchy above, we only desired our X to be metrizable.

In reality, the Borel hierarchy is usually studied specifically on Polish spaces. If X is Polish, the hierarchy is proper, i.e. all the classes are distinct if X is uncountable.

A more formal way to express these ideas is presented in the theorem below.

Theorem 6 (Non-collapse). [3, p. 168] *Suppose X is an uncountable Polish space. Then, $\forall \xi$ $\Sigma_\xi^0 \neq \Pi_\xi^0$. And hence, $\Delta_\xi^0 \subsetneq \Sigma_\xi^0 \subsetneq \Delta_{\xi+1}^0$. Similarly for Π_ξ^0 .*

Basically, we indeed have a hierarchy, which doesn't "collapse" - "higher" levels contain elements which are not present in the lower levels.

The reader may wonder¹⁸ whether this is meaningful - do there even exist sets, whose Borel hierarchy “collapses”? And indeed, there are such sets. Once again, take $\mathbb{Q}$ with the discrete topology. Then every subset of X is open (and also closed), and so the hierarchy immediately collapses, i.e. $\forall \xi \Sigma_\xi^0 = \Pi_\xi^0 = \Sigma_{\xi+1}^0$.

The full proof of this theorem requires its own separate block of theory [3, pp. 167-168], so here we will instead look at the intuition and outline of why it is true.

Since X is uncountable and Polish, the Cantor set C can be embedded into X . Hence, it suffices to show [3, p. 168] that $\Sigma_\xi^0(C) \neq \Pi_\xi^0(C)$.

Let us take a set¹⁹ $U \subset C \times C$ which belongs to $\Sigma_\xi^0(C \times C)$, and “encodes” all members of $\Sigma_\xi^0(C)$, such that as y ranges over C , $U_y = \{x \in C : (y, x) \in U\}$ ranges over $\Sigma_\xi^0(C)$. Next, we assume for contradiction $\Sigma_\xi^0(C) = \Pi_\xi^0(C)$. The key trick of this proof is pretty similar to the trick in Russell’s paradox. Let’s construct $A = \{y \in C : (y, y) \notin U\}$, which basically consists of those y which don’t belong to the set they encode. Some subtle manipulation shows that $A \in \Pi_\xi^0(C)$, and by our assumption, $A \in \Sigma_\xi^0(C)$. But then we have $A = U_{y_0}$ for some y_0 , and asking whether y_0 is in A gives us a contradiction [3, p. 168].

Hence, Polish spaces give us this beautiful reassurance that the hierarchy is indeed meaningful and it makes sense to study. It is worth taking a moment to appreciate how the properties we looked at the very beginning of this essay work together to provide the hierarchy with richness and depth.

4 Conclusion

In this essay, we looked at two staples of classical Descriptive Set theory - Polish spaces and the Borel hierarchy. Starting with seemingly simple requirements for our spaces, we proved a fascinating variety of properties: their subsets are Polish precisely when they’re G_δ , they embed into the Hilbert cube, and, if they’re compact, they are continuous images of the Cantor set. Even more, we saw a beautiful interplay of infinite games from Game theory and Set theory.

¹⁸One may also wonder what is Σ_ξ^1 (0 replaced by 1) - these are analytic sets, definitely out of scope of this essay, but it seems valuable to mention that there are (as usual) concepts a level of complexity above what we’ve seen so far.

¹⁹The fact that it’s possible to construct a set with the desired properties is a theorem of its own [3, p. 168] and is taken for granted in this essay. However, it is important to point out that the existence is guaranteed because $C \times C$ is separable metrizable. The curious readers should have a look at *universal sets*.

A recurring theme throughout the essay was the “niceness” of Polish spaces, which was perfectly exemplified by the non-collapse property of the Borel Hierarchy, a way to organize subsets of a metrizable space into an insightful pattern. This is essentially what Descriptive Set theory is about – spaces that are simple at first glance end up being insightful on their own, but also end up being the foundation of powerful configurations.

In this essay, we just touched upon the very tip of the iceberg – there are projective and analytic sets that form their own hierarchy, a whole separate area of Baire category theory, and many questions that remain unanswered to this day. Nevertheless, hopefully the reader now has a better understanding of why Polish spaces deserve the attention they’re getting, and has an appetite to discover more about the theory at the entry point to which Polish spaces stand.

References

- [1] Ryszard Engelking. *General Topology*. Heldermann, 1989.
- [2] Michael C. Gemignani. *Elementary Topology*. Addison-Wesley Publishing Company, second edition, 1971.
- [3] Alexander S. Kechris. *Classical Descriptive Set Theory*. Springer, 1995.
- [4] John L. Kelley. *General Topology*. Springer, 1955.
- [5] Ivan Khatchaturian. 12. metric spaces and metrizability. University faculty website, 2018. URL <https://www.math.toronto.edu/ivan/mat327/docs/notes/12-metric.pdf>. Lecture notes, MAT327 Topology, University of Toronto.
- [6] Jim Kuzmanovich. Polish school of mathematics, 2000. URL <https://users.wfu.edu/kuz/Stamps/PolishSchool/PolishSchool.htm>. Article on the faculty of Wake Forest University website.
- [7] James R. Munkres. *Topology*. Prentice Hall, Inc., second edition, 2000.
- [8] Joseph Pomianowski. Mathematics and polish national identity. *Cosmopolitan Review*, 3(1), March 2011.
- [9] Julian P. Revalski. The banach-mazur game: History and recent developments, 2004. URL http://www1.univ-ag.fr/aoc/activite/revalski/Banach-Mazur_Game.pdf.

- [10] René L. Schilling and Franziska Kühn. *Counterexamples in Measure and Integration*. Cambridge University Press, 2021.
- [11] Leen A. Steen and J. Arthur Seebach. *Counterexamples in Topology*. Holt, Rinehart and Winston, Inc., 1970.